Tarea 1 Mecánica Estadística Avanzada

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April 12, 2026

Como primera tarea les corresponde realizar 10 de los 21 problemas 1-23, 1-24, 1-27, 1-28, 1-30, 2-13, 2-15, 2-16, 2-17, 2-19, 3-4, 3-5, 3-7, 3-8, 3-9, 3-17, 3-19, 3-20, 3-22, 3-23, 3-24 del libro de Mc. Quarrie.

1 Problema 1.23

Verify the calculation:

$$\Omega(E, \Delta E) = \frac{1}{\Gamma(N+1)\Gamma(3N/2)} \left(\frac{2\pi m a^2}{h^2}\right)^{3N/2} E^{(3N/2-1)} \Delta E \tag{1.1}$$

In this case, $E = 3NkT/2$. If we take $T = 300°\text{K}, m = 10^{-22}$ g, $a = 10$ cm, $N = 6.02 \times 10^{23}$, and ΔE equal to $0.01E$, we get $\Omega(E, \Delta E)$ to be $O\left(10^N\right)$.

Respuesta: Resumiendo los valores dados:

$$\begin{cases} E &= 3NkT/2 \\ T &= 300 \text{ K} \\ m &= 10^{-22} \text{ g} &= 10^{-25} \text{ kg} \\ a &= 10 \text{ cm} &= 0.1 \text{ m} \\ N &= 6.02 \times 10^{23} \\ \Delta E &= 0.01E &= E/100 \end{cases} \tag{1.2}$$

Luego,

$$\begin{aligned} \Omega(E, \Delta E) &= \frac{1}{\Gamma(N+1)\Gamma(3N/2)} \left(\frac{2\pi m a^2}{h^2}\right)^{3N/2} \frac{E^{(3N/2-1)} \cdot E}{100} \\ &= \frac{1}{100} \frac{1}{\Gamma(N+1)\Gamma(3N/2)} \left(1.43109 \times 10^{40} E\right)^{3N/2} \\ &= \frac{1}{100} \frac{1}{\Gamma(N+1)\Gamma(3N/2)} \left(8.38389 \times 10^{29}\right)^N N^{3N/2} \end{aligned}$$

Usando la aproximación de Stirling:

$$\Gamma(x+1) \approx \sqrt{2\pi x} \left(\frac{x}{e}\right)^x \tag{1.3}$$

$$\Rightarrow \Gamma(N+1)\Gamma(3N/2) \approx 2\pi\sqrt{N}\sqrt{3N/2-1}\left(\frac{N}{e}\right)^N\left(\frac{3N/2-1}{e}\right)^{3N/2-1}$$

Pero $3N/2-1 \approx 3N/2$, entonces

$$\begin{aligned}
\Rightarrow \Gamma(N+1)\Gamma(3N/2) &\approx 2\pi\sqrt{N}\sqrt{3N/2}\left(\frac{N}{e}\right)^N\left(\frac{3N/2-1}{e}\right)^{3N/2-1} \\
&\approx \sqrt{6}\pi\frac{N^{N+1}(3N/2)^{3N/2}}{e^{5N/2}}\frac{2e}{3N} \\
&\approx \frac{2\sqrt{6}\pi e}{3}\frac{3^{3N/2}}{2^{3N/2}e^{5N/2}}N^N N^{3N/2} \\
&\approx \frac{2\sqrt{6}\pi e}{3}\left(\frac{3^3}{2^3e^5}\right)^{N/2}N^{5N/2} \\
&\approx \frac{2\sqrt{6}\pi e}{3}\left(0.1508\right)^N N^{5N/2}
\end{aligned}$$

Luego

$$\begin{aligned}
\Omega(E,\Delta E) &\approx \frac{1}{100}\frac{3}{2\sqrt{6}\pi e\left(0.1508\right)^N N^{5N/2}}\left(8.38389\times 10^{29}\right)^N N^{3N/2} \\
&\approx \underbrace{\frac{3}{200\sqrt{6}\pi e}}_{\approx 0}\left(5.55962\times 10^{30}\right)^N N^{-N} \\
&\approx \left(\frac{5.55962\times 10^{30}}{N}\right)^N \\
&\approx \left(9.23524\times 10^6\right)^N \propto 10^{6N}
\end{aligned}$$

Pero $6N \propto N$, entonces

$$\Omega(E,\Delta E) \propto 10^N \tag{1.4}$$

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2 Problema 1.24

We need to know the volume of an N-dimensional sphere in order to derive Eq. (1-36). This can be determined by the following device. Consider the integral

$$I = \int_{-\infty}^{\infty}\int e^{-\left(x_1^2+x_2^2+\cdots+x_N^2\right)}dx_1dx_2\cdots dx_N \tag{2.1}$$

a. First show that $I = \pi^{N/2}$.

Respuesta: De la ec. 2.1

$$
\begin{aligned}
I &= \int_{-\infty}^{\infty} \int e^{-\left(x_1^2+x_2^2+\cdots+x_N^2\right)} dx_1 dx_2 \cdots dx_N \\
&= \int_{-\infty}^{\infty} \int e^{-x_1^2} e^{-x_2^2} \cdots e^{-x_N^2} dx_1 dx_2 \cdots dx_N \\
&= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^N \\
&= \left(\sqrt{\pi}\right)^N = \pi^{N/2}
\end{aligned} \tag{2.2}
$$

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b. Now one can formally transform the volume element $dx_1 dx_2 \cdots dx_N$ to N-dimensional spherical (hyperspherical) coordinates to get

$$
\int_{\text{angles}} dx_1 dx_2 \cdots dx_N \to r^{N-1} S_N dr \tag{2.3}
$$

where S_N is the factor that arises upon integration over the angles. Show that $S_2 = 2\pi$ and $S_3 = 4\pi$.

Respuesta: Para $N = 2$

$$
\iint_{\text{angles}} dx_1 dx_2 \tag{2.4}
$$

De donde podemos identificar dx_1, dx_2 con las coordenadas polares:

$$
\begin{aligned}
dx_1 = dr \quad & \begin{cases} r_{\text{i}} &= 0 \\ r_{\text{f}} &= R \end{cases} \\
dx_2 = r d\theta \quad & \begin{cases} \theta_{\text{i}} &= 0 \\ \theta_{\text{f}} &= 2\pi \end{cases}
\end{aligned}
$$

Luego, ec. 2.4

$$
\begin{aligned}
\iint_{\text{angles}} dx_1 dx_2 &= \iint_0^{2\pi} r d\theta dr \\
&= \int \underbrace{2\pi}_{S_2} \cdot \underbrace{r}_{r^{N-1}} dr
\end{aligned} \tag{2.5}
$$

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Para $N = 3$

$$
\iiint_{\text{angles}} dx_1 dx_2 dx_3 \tag{2.6}
$$

De donde podemos identificar dx_1, dx_2, dx_3 con las coordenadas esféricas:

$$dx_1 = dr \qquad \begin{cases} r_{\mathrm{i}} &= 0 \\ r_{\mathrm{f}} &= R \end{cases}$$

$$dx_2 = r d\theta \qquad \begin{cases} \theta_{\mathrm{i}} &= 0 \\ \theta_{\mathrm{f}} &= \pi \end{cases}$$

$$dx_3 = r \sin\theta d\varphi \begin{cases} \varphi_{\mathrm{i}} &= 0 \\ \varphi_{\mathrm{f}} &= 2\pi \end{cases}$$

Luego, ec. 2.6

$$\begin{aligned} \iiint_{\text{angles}} dx_1 dx_2 &= \iint_0^{2\pi} \int_0^{\pi} r \cdot r \sin\theta d\theta d\varphi dr \\ &= \iint_0^{2\pi} 2r^2 d\varphi dr \\ &= \int \underbrace{4\pi}_{S_3} \cdot \underbrace{r^2}_{r^{N-1}} dr \end{aligned} \tag{2.7}$$

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c. S_N can be determined for any N by writing I in hyperspherical coordinates:

$$I = \int_0^{\infty} e^{-r^2} r^{N-1} S_N dr \tag{2.8}$$

Show that $I = S_N \Gamma(N/2)/2$, where $\Gamma(x)$ is the gamma function (see Problem 1-58).

Respuesta: La definición de la función Gamma:

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx \tag{2.9}$$

El cuál, para cualquier entero positivo n, es igual a

$$\Gamma(n) = (n-1)! \tag{2.10}$$

Y tiene la propiedad fundamental:

$$\Gamma(z+1) = z\Gamma(z) \tag{2.11}$$

De la ec. 2.8:

$$I = S_N \int_0^{\infty} r^{N-1} e^{-r^2} dr$$

Con el cambio de variables:

$$u \equiv r^2, \quad du = 2r dr$$

Obtenemos:

$$
\begin{aligned}
I &= S_N \int_0^\infty u^{\frac{N-1}{2}} e^{-u} dr \\
&= S_N \int_0^\infty u^{N/2-1} \sqrt{u} \cdot e^{-u} dr \\
&= S_N \frac{1}{2} \int_0^\infty u^{N/2-1} \cdot e^{-u} \underbrace{2rdr}_{du} \\
&= \frac{S_N}{2} \int_0^\infty u^{N/2-1} e^{-u} du \\
I &= \frac{S_N}{2} \Gamma(N/2) \qquad (2.12)
\end{aligned}
$$

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d. Equate these two values (eqs. 2.2 and 2.12) for I to get

$$
S_N = \frac{2\pi^{N/2}}{\Gamma(N/2)} \qquad (2.13)
$$

Respuesta: Tomando las ecs. 2.2 y 2.12:

$$
\begin{aligned}
I = \pi^{N/2} &= \frac{S_N}{2} \Gamma(N/2) \\
\Rightarrow S_N &= \frac{2\pi^{N/2}}{\Gamma(N/2)}
\end{aligned}
$$

■

e. Show that this reduces correctly for $N = 2$ and 3.

Respuesta: Para $N = 2$:

$$
\begin{aligned}
S_2 &= \frac{2\pi^{2/2}}{\Gamma(2/2)} \\
&= \frac{2\pi}{\Gamma(1)} = \frac{2\pi}{0!} \\
&= 2\pi
\end{aligned}
$$

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Para $N = 3$, y considerando que $\Gamma(3/2) = \sqrt{\pi}/2$:

$$
\begin{aligned}
S_3 &= \frac{2\pi^{3/2}}{\Gamma(3/2)} \\
&= \frac{2\pi^{3/2}}{\pi^{1/2}/2} \\
&= 4\pi
\end{aligned}
$$

■

f. Lastly now, convince yourself that the volume of an N-dimensional sphere of radius a is given by

$$V_N = \int_0^a S_N r^{N-1} dr = \frac{\pi^{N/2}}{\Gamma\left(\frac{N}{2}+1\right)} a^N$$

and show that this reduces correctly for $N = 2$ and 3.

Respuesta: Tomando

$$\begin{aligned} V_N &= \int_0^a S_N r^{N-1} dr \\ &= S_N \int_0^a r^{N-1} dr \\ &= \frac{2\pi^{N/2}}{\Gamma(N/2)} \left[\frac{r^N}{N}\right]_0^a \\ &= \frac{2\pi^{N/2}}{\Gamma(N/2)} \frac{a^N}{N} \end{aligned}$$

Pero, usando la propiedad 2.11, $\Gamma\left(\frac{N}{2}+1\right) = \frac{N}{2}\Gamma\left(\frac{N}{2}\right)$, obtenemos

$$\begin{aligned} V_N &= \pi^{N/2} a^N \frac{2}{N\Gamma(N/2)} \\ &= \frac{\pi^{N/2}}{\Gamma(N/2+1)} a^N \end{aligned}$$

■

3 Problema 1.27

Show for an isothermal process that $w_{\text{rev}} > w_{\text{irrev}}$ and $q_{\text{rev}} > q_{\text{irrev}}$.

Respuesta: De la Primera Ley de la Termodinámica:

$$dE = \delta Q - \delta W \tag{3.1}$$

Y la Segunda Ley de la Termodinámica

$$\delta Q \leq \frac{dS}{T} \tag{3.2}$$

Con la igualdad valiendo **solo** para procesos reversibles. Obtenemos

$$\begin{aligned} \delta W &= \delta Q - dE \\ \delta W &\leq \frac{dS}{T} - dE \end{aligned}$$

Ahora, para los casos reversibles e irreversibles:

$$\begin{aligned}\delta W_{\text{rev}} &= \frac{dS}{T} - dE \\ \delta W_{\text{irrev}} &< \frac{dS}{T} - dE\end{aligned}$$

Luego

$$\delta W_{\text{rev}} > \delta W_{\text{irrev}}$$

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Además, . 3.2

$$\begin{aligned}\delta Q_{\text{rev}} &= \frac{dS}{T} \\ \delta Q_{\text{irrev}} &< \frac{dS}{T}\end{aligned}$$

Then,

$$\delta Q_{\text{rev}} > \delta Q_{\text{irrev}}$$

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4 Problema 1.28

Derive the thermodynamic equation

$$C_p - C_V = \left[p + \left(\frac{\partial E}{\partial V}\right)_T\right]\left(\frac{\partial V}{\partial T}\right)_p \tag{4.1}$$

and evaluate this difference for an ideal gas and a gas that obeys the van der Waals equation.

Respuesta: Las definiciones de C_V y C_p:

$$C_V = \left(\frac{\partial E}{\partial T}\right)_V, \qquad C_p = \left(\frac{\partial E}{\partial T}\right)_p + p\left(\frac{\partial V}{\partial T}\right)_p \tag{4.2}$$

Ahora, la función de estado de la energía escrita en términos de T, V:

$$\begin{aligned}dE &= \left(\frac{\partial E}{\partial T}\right)_V dT + \left(\frac{\partial E}{\partial V}\right)_T dV \\ \Rightarrow \left(\frac{\partial E}{\partial T}\right)_p &= \left(\frac{\partial E}{\partial T}\right)_V + \left(\frac{\partial E}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \\ &= C_V + \left(\frac{\partial E}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p\end{aligned}$$

Sustituyendo esto en la expresión para C_p en la ec. 4.2

$$C_p = C_V + \left(\frac{\partial E}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p + p\left(\frac{\partial V}{\partial T}\right)_p$$

$$\Rightarrow C_p - C_V = \left[\left(\frac{\partial E}{\partial V}\right)_T + p\right]\left(\frac{\partial V}{\partial T}\right)_p \tag{4.3}$$

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Para evaluar la ec. 4.3 para cualquier gas ideal o de Van der Waals, debemos encontrar las expresiones para

$$\left(\frac{\partial E}{\partial V}\right)_T \text{ and } \left(\frac{\partial V}{\partial T}\right)_p$$

Tomando la ecuación fundamental para $E(S, V)$:

$$dE = TdS - pdV \tag{4.4}$$

$$\left(\frac{\partial E}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - p$$

Pero la relación de Maxwell

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V \tag{4.5}$$

Se convierte en

$$\left(\frac{\partial E}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p \tag{4.6}$$

En el caso del gas ideal, tomamos la ley de los gases ideales:

$$pV = NkT \tag{4.7}$$

$$p = \frac{NkT}{V}$$

Obtenemos inmediatamente

$$V = \frac{NkT}{p}$$

$$\Rightarrow \left(\frac{\partial V}{\partial T}\right)_p = \frac{Nk}{p} \tag{4.8}$$

Y usando ec. 4.6

$$\left(\frac{\partial E}{\partial V}\right)_T = T\frac{Nk}{V} - \frac{NkT}{V}$$

$$\left(\frac{\partial E}{\partial V}\right)_T = 0 \tag{4.9}$$

Utilizando las ecs. 4.8 y 4.9 en la ec. 4.3

$$C_p - C_V = [0 + p]\,\frac{Nk}{p}$$
$$C_p - C_V = Nk \tag{4.10}$$

■

La ecuación de Van der Waals se define:

$$p = \frac{NkT}{V - Nb} - a\frac{N^2}{V^2} \tag{4.11}$$

De aquí, obtenemos

$$\left(\frac{\partial p}{\partial T}\right)_V = \frac{Nk}{V - Nb} \tag{4.12}$$

Utilizando las ecs. 4.11 y 4.12 en la ec. 4.6

$$\left(\frac{\partial E}{\partial V}\right)_T = T\frac{Nk}{V - Nb} - \left(\frac{NkT}{V - Nb} - a\frac{N^2}{V^2}\right)$$
$$\left(\frac{\partial E}{\partial V}\right)_T = a\frac{N^2}{V^2} \tag{4.13}$$

Para la segunda derivada, reordenamos y diferenciamos la ec. 4.11

$$NkT = \left(p + a\frac{N^2}{V^2}\right)(V - Nb)$$
$$\Rightarrow Nk = \frac{\partial}{\partial T}\left[\left(p + a\frac{N^2}{V^2}\right)(V - Nb)\right]_p$$
$$Nk = (V - Nb)\,\frac{\partial}{\partial T}\left[\left(p + a\frac{N^2}{V^2}\right)\right]_p + \left(p + a\frac{N^2}{V^2}\right)\frac{\partial}{\partial T}\left[(V - Nb)\right]_p$$
$$Nk = -\,(V - Nb)\left(2a\frac{N^2}{V^3}\right)\left(\frac{\partial V}{\partial T}\right)_p + \left(p + a\frac{N^2}{V^2}\right)\left(\frac{\partial V}{\partial T}\right)_p$$
$$\Rightarrow \left(\frac{\partial V}{\partial T}\right)_p = \frac{Nk}{\left(p + a\frac{N^2}{V^2}\right) - (V - Nb)\left(2a\frac{N^2}{V^3}\right)} \tag{4.14}$$

Utilizando las ecs. 4.13 y 4.14 dentro de la ec. 4.3

$$C_p - C_V = \frac{Nk\cancel{\left(a\frac{N^2}{V^2} + p\right)}}{\cancel{\left(p + a\frac{N^2}{V^2}\right)} - (V - Nb)\left(2a\frac{N^2}{V^3}\right)}$$
$$= \frac{Nk}{1 - \dfrac{(V - Nb)\left(2a\frac{N^2}{V^3}\right)}{\left(a\frac{N^2}{V^2} + p\right)}}$$

Pero desde la ec. 4.11 sabemos que $a\dfrac{N^2}{V^2} + p = NkT/(V - Nb)$:

$$C_p - C_V = \frac{Nk}{1 - \dfrac{(V - Nb)^2 \left(2aN^2/V^3\right)}{NkT}} \tag{4.15}$$

■

5 Problema 2.13

Show that for a particle confined to a cube of length a that

$$p_j = \frac{2}{3}\frac{E_j}{V}$$

By taking the ensemble average of both sides, we have

$$\bar{p} = \frac{2}{3}\frac{\bar{E}}{V}$$

If we use the fact that $E = \frac{3}{2}NkT$ (to be proved in Chapter 5), we get the ideal gas equation of state.

Respuesta: Sabemos que para un cubo de lado a, tendremos:

$$V = a^3 \tag{5.1}$$

De la ecuación fundamental para $E(S, V, N)$

$$dE = TdS - pdV + \mu dN \tag{5.2}$$

$$\Rightarrow p = -\left(\frac{\partial E}{\partial V}\right)_{S,N} \tag{5.3}$$

$$= -\left(\frac{\partial E}{\partial a}\right)_{S,N}\left(\frac{\partial a}{\partial V}\right)_{S,N}$$

$$= -\frac{1}{3a^2}\left(\frac{\partial E}{\partial a}\right)_{S,N} \tag{5.4}$$

Y la energía vendrá de los valores propios del operador de energía:

$$E_j = \frac{\pi^2\hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right) \tag{5.5}$$

Luego, usando la ec. 5.4

$$\begin{aligned} p &= -\frac{1}{3a^2}\frac{\partial}{\partial a}\left[\frac{\pi^2\hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)\right]_{S,N} \\ &= -\frac{1}{3a^2}\left[-\frac{\pi^2\hbar^2}{ma^3}\left(n_x^2 + n_y^2 + n_z^2\right)\right] \\ &= \frac{2}{3a^3}\cdot\underbrace{\frac{\pi^2\hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)}_{E_j} \end{aligned}$$

Luego finalmente:

$$p = \frac{2}{3V} E_j \tag{5.6}$$

■

En el caso cuántico, tendremos el valor esperado:

$$\langle A \rangle = \sum_i P_i A_i \tag{5.7}$$

Donde A_i es algún observable y P_i es la probabilidad asociada, para la presión:

$$\begin{aligned} \langle p \rangle &= \sum_j P_j p_j \\ &= \frac{2}{3V} \sum_j P_j E_j \\ &= \frac{2}{3V} \bar{E} \end{aligned}$$

Luego, usando $E = \frac{1}{2} NkT$ tenemos

$$\begin{aligned} \bar{p} &= \frac{2}{3V} \frac{3}{2} NkT \\ \bar{p}V &= NkT \end{aligned} \tag{5.8}$$

■

6 Problema 2.15

In Chapter 11 we shall approximate the partition function of a crystal by

$$Q = \left(\frac{e^{-h\nu/2kT}}{1 - e^{-h\nu/kT}} \right)^{3N} e^{U_0/kT} \tag{6.1}$$

where $h\nu/k \equiv \Theta_E$ is a constant characteristic of the crystal, and U_0 is the sublimation energy of the crystal. Calculate the heat capacity from this simple partition function and show that at high temperatures, one obtains the law of Dulong and Petit, namely, that $C_V \to 3Nk$ as $T \to \infty$.

Respuesta: Sabemos que

$$C_V = \left(\frac{\partial U}{\partial T} \right)_{V,N} \tag{6.2}$$

Y

$$U = kT^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{V,N} \tag{6.3}$$

Utilizando la ec. 6.1 dentro de la ec. 6.3 obtenemos:

$$U = \frac{3}{2} Nh\nu \coth \left(\frac{h\nu}{2kT} \right) - U_0 \tag{6.4}$$

Luego, usando ec. 6.2

$$C_V = \frac{3Nh^2\nu^2}{4kT^2}\text{csch}^2\left(\frac{h\nu}{2kT}\right) \tag{6.5}$$

■

En el límite $T \to \infty$:

$$C_V = 3Nk + \mathcal{O}\left(\frac{1}{T^2}\right) \tag{6.6}$$

■

NOTA: Se usó Wolfram Mathematica para las derivadas.

7 Problema 2.16

In Chapter 13 of this author's textbook Statistical Thermodynamics, it is shown that the partition function of an ideal gas of diatomic molecules in an external electric field $\mathscr{E}$ is

$$Q(N, V, T, \mathscr{E}) = \frac{[q(V, T, \mathscr{E})]^N}{N!} \tag{7.1}$$

where

$$q(V, T, \mathscr{E}) = V\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\left(\frac{8\pi^2 IkT}{h^2}\right)\frac{e^{-h\nu/2kT}}{\left(1 - e^{-h\nu/kT}\right)}\left(\frac{kT}{\mu\mathscr{E}}\right)\sinh\left(\frac{\mu\mathscr{E}}{kT}\right) \tag{7.2}$$

Here I is the moment of inertia of the molecule; ν is its fundamental vibrational frequency; and μ is its dipole moment. Using this partition function along with the thermodynamic relation,

$$dF = -SdT - pdV - Md\mathscr{E} \tag{7.3}$$

where $M = N\bar{\mu}$, where $\bar{\mu}$ is the average dipole moment of a molecule in the direction of the external field $\mathscr{E}$, show that

$$\bar{\mu} = \mu\left[\coth\left(\frac{\mu\mathscr{E}}{kT}\right) - \frac{kT}{\mu\mathscr{E}}\right] \tag{7.4}$$

Sketch this result versus $\mathscr{E}$ from $\mathscr{E} = 0$ to $\mathscr{E} = \infty$ and interpret it.

Respuesta: Tomando la ec. 7.3:

$$dF = -SdT - pdV - Md\mathscr{E}$$

$$\Rightarrow \left(\frac{\partial F}{\partial \mathscr{E}}\right)_{T,V} = -M \tag{7.5}$$

La energía libre de Helmholtz se relaciona con la función de partición a través de:

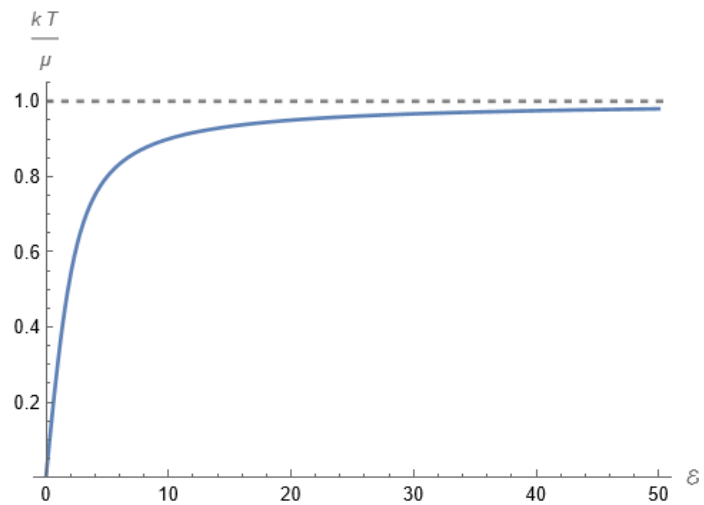
$$F = -kT\ln Q \tag{7.6}$$

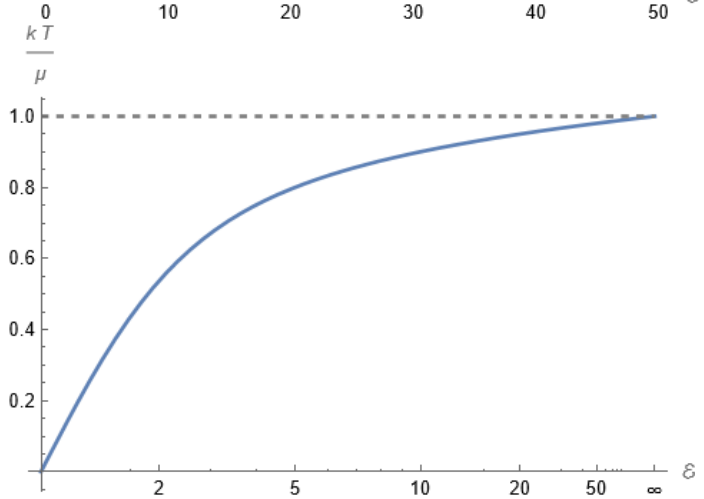
Por lo que usando ec. 7.6 en 7.5:

$$
\begin{aligned}
M &= kT\left(\frac{\partial \ln Q}{\partial \mathscr{E}}\right)_{T,V} \\
N\bar{\mu} &= -\frac{NkT}{\mathscr{E}} + \coth\left(\frac{\mathscr{E}\mu}{kT}\right) \\
N\bar{\mu} &= N\mu\left[\coth\left(\frac{\mathscr{E}\mu}{kT}\right) - \frac{kT}{\mu\mathscr{E}}\right] \\
\bar{\mu} &= \mu\left[\coth\left(\frac{\mathscr{E}\mu}{kT}\right) - \frac{kT}{\mu\mathscr{E}}\right] \qquad (7.7)
\end{aligned}
$$

■

Graficando:





NOTA: Se usó Wolfram Mathematica para las derivadas y los gráficos.

8 Problema 2.17

In Chapter 14 we shall derive an approximate partition function for a dense gas, which is of the form

$$Q(N,V,T) = \frac{1}{N!}\left(\frac{2\pi mkT}{h^2}\right)^{3N/2}(V-Nb)^N e^{aN^2/VkT} \tag{8.1}$$

where a and b are constants that are given in terms of molecular parameters. Calculate the equation of state from this partition function. What equation of state is this? Calculate the thermodynamic energy and the heat capacity and compare it to Problem 1-30.

Respuesta: Introduciendo la ec. 8.1 en la relación

$$p = kT\left(\frac{\partial \ln Q}{\partial V}\right)_{T,N} \tag{8.2}$$

obtenemos

$$p = \frac{NkT}{V-Nb} - \frac{aN^2}{V^2} \tag{8.3}$$

La cual coincide con la ecuación de de Van der Waals (ec. 4.11). ■

Luego, para la energía y la capacidad calórica, aplicamos las ecs. 6.3 y 6.2:

$$U = kT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{V,N}$$

$$U = \frac{3NkT}{2} - \frac{aN^2}{V} \tag{8.4}$$

■

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{V,N}$$

$$C_V = \frac{3Nk}{2} \tag{8.5}$$

■

NOTA: Se usó Wolfram Mathematica para las derivadas.

9 Problema 3.4

State and use Euler's theorem to show

$$p = kT\left(\frac{\partial \ln \Xi}{\partial V}\right)_{\mu,T} = kT\frac{\ln \Xi}{V} \tag{9.1}$$

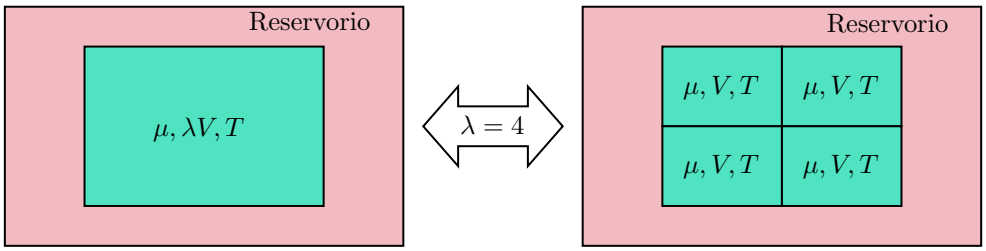
Respuesta: El teorema de Euler dice que para cualquier función f homogénea de orden n, es decir, que cumpla con

$$f(\lambda x_1, \lambda x_2, \ldots, \lambda x_N) = \lambda^n f(x_1, x_2, \ldots, x_N)$$

entonces

$$nf(x_1, x_2, \ldots, x_N) = x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} + \cdots + x_N \frac{\partial f}{\partial x_N} \tag{9.2}$$

Si consideramos un sistema de volumen V en equilibrio con un reservorio, y aumentamos el volumen por un factor λ, manteniendo fijos μ, T, tendremos, equivalentemente, λ copias idénticas e independientes una al lado de la otra. Escalar el sistema de esta forma nos deja con una función



$\Xi(\mu, \lambda V, T)$ $\qquad$ $[\Xi(\mu, V, T)]^\lambda$

de partición igual al producto de las funciones de los subsistemas, es decir:

$$\Xi(\mu, \lambda V, T) = [\Xi(\mu, V, T)]\, \lambda \tag{9.3}$$

Luego:

$$\ln \Xi(\mu, \lambda V, T) = \lambda \ln \Xi(\mu, V, T) \tag{9.4}$$

Por lo que $\ln \Xi$ es homogénea de orden 1 respecto a V, mientras que μ, T no escalan por ser intensivas. Entonces, siguiendo la ec. 9.2:

$$\begin{aligned} \ln \Xi(\mu, V, T) &= V \left(\frac{\partial \ln \Xi}{\partial V} \right)_{\mu, T} \\ \Rightarrow \frac{\ln \Xi}{V} &= \left(\frac{\partial \ln \Xi}{\partial V} \right)_{\mu, T} \end{aligned} \tag{9.5}$$

Aplicando la relación:

$$p = kT \frac{\ln \Xi}{V} \tag{9.6}$$

Obtenemos finalmente:

$$p = kT \frac{\ln \Xi}{V} = kT \left(\frac{\partial \ln \Xi}{\partial V} \right)_{\mu, T} \tag{9.7}$$

■

10 Problema 3.5

Show that the entropy given by

$$S = \frac{\bar{E}}{T} - \frac{\bar{N}\mu}{T} + k \ln \Xi \tag{10.1}$$

goes to zero as T goes to zero.

Respuesta: Considerando

$$\Xi = \sum_{N,j} e^{-(E_{N,j} - N\mu)/kT} \tag{10.2}$$

Definiendo

$$G_{N,j} \equiv E_{N,j} - N\mu \tag{10.3}$$

Y tomando el límite $T \to 0$

$$\lim_{T \to 0} \Xi = \sum_{N,j} e^{-G_{N,j}/kT}$$

Tendremos que este estará dominado por el valor mínimo de $G_{N,j}$, o el *ground state*:

$$G_0 = \min\left(G_{N,j}\right) \tag{10.4}$$

Factorizando:

$$\begin{aligned}
\lim_{T \to 0} \Xi &= e^{-G_0/kT} \left[\sum_{N,j} e^{-(G_{N,j} - G_0)/kT} \right] \\
&= e^{-G_0/kT} \left[\sum_{G_{N,j} = G_0} \underbrace{e^{-(G_0 - G_0)/kT}}_{1} + \sum_{G_{N,j} > G_0} e^{-(G_{N,j} - G_0)/kT} \right] \\
&= e^{-G_0/kT} \left[g_0 + \sum_{G_{N,j} > G_0} e^{-(G_{N,j} - G_0)/kT} \right]
\end{aligned}$$

Donde g_0 es la degeneración de G_0, es decir, la cantidad de estados que comparten este estado de energía. Evaluando el límite tendremos que los términos de orden superior se anularán:

$$\Xi \approx e^{-G_0/kT} \left[g_0 + 0\right] = g_0 e^{-G_0/kT} \tag{10.5}$$

Luego,

$$\ln \Xi \approx \ln g_0 - \frac{G_0}{kT} \tag{10.6}$$

Reemplazando esto en la ec. 10.1:

$$\begin{aligned}
S &\approx \frac{\bar{E} - \bar{N}\mu}{T} + k \left(\ln g_0 - \frac{G_0}{kT} \right) \\
&\approx \frac{\bar{E} - \bar{N}\mu}{T} + k \ln g_0 - \frac{G_0}{T}
\end{aligned}$$

Pero tendremos que, cuando $T \to 0$, los valores promedio $\bar{E}$ y $\bar{N}$ tienden a los correspondientes al *ground state*, por ser este el estado más probable. Por lo que, usando la definición de 10.3:

$$\lim_{T \to 0} \left(\bar{E} - \bar{N}\mu \right) \approx G_0 \tag{10.7}$$

Luego

$$\begin{aligned} S &\approx \frac{G_0}{T} + k \ln g_0 - \frac{G_0}{T} \\ S &\approx k \ln g_0 \end{aligned} \tag{10.8}$$

Finalmente, la Tercera Ley de la Termodinámica nos dice que, para la mayoría de los sistemas cristalinos, solo existirá un sistema en el estado base:

$$g_0 = 1 \tag{10.9}$$

Por lo que:

$$S \approx k \ln 1 = 0 \tag{10.10}$$

■